

The all-order expansion of the one-dimensional chart integral

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Abstract

For a chart amplitude $\eta(u)e^{\beta s \xi(u)}$ with ξ, η admitting order- R Taylor expansions at $u = 0$ on $[0, b]$ and ξ Lipschitz at 0, the scaled integral $\int_0^b u^h e^{-\beta(Nu^k)^2} \eta(u) e^{\beta(Nu^k)\xi(u)} du$ equals $\sum_{j < R} c_j N^{-(p+j/k)} + O(N^{-(p+R/k)})$ with $p = (h+1)/k$ and

$$c_j = \frac{1}{k} \sum_{m < R} \frac{\beta^m}{m!} e_{j,m} S_{p+j/k+m}(x_0), \quad e_{j,m} = \sum_{\ell \leq j} y_\ell [u^{j-\ell}](g_R^m),$$

$$S_\lambda(a) = \int_0^\infty s^{\lambda-1} e^{-\beta s^2 + \beta a s} ds.$$

This assembles the abstract kernel-jet transfer (unit 92) with the amplitude jet remainder (unit 95): it is the all-order form of the one-dimensional case of `thm:TaylorTree`, with every coefficient an explicit finite combination of the paper's S_λ functions. Zero `sorry/axiom`.

1 The things formalised

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theorem moment_integrableOn_of_envelope ( a' A : ) (h : 0 < ) (h : -1 < ) (j R : )
  (c : → ) (hc : Measurable c)
  (hbound : s, 0 < s → |c s| A * (1 + s) ^ R * Real.exp ( * s * a')) :
  IntegrableOn (fun s => s ^ * (1 + |Real.log s|) ^ j * (Real.exp (-
    * s ^ 2) * |c s|))
  (Ioi 0)
theorem ampJet_abs_le ... : |ampJet x y R j s|
  ( range (j + 1), |y| * expJetConst x R (j - )) * (1 + s) ^ R
noncomputable def chartJet ( : ) (x y : → ) (R j : ) (s : ) : :=
  Real.exp ( * s * x 0) * ampJet x y R j s
theorem oneDChart_all_orders ( b K L p : ) (h k R : ) (h : 0 < ) (hb : 0 < b)
  (hK : 0 K) (hL : 0 L) (hk : 0 < k) (hp : ((h : ) + 1) / k = p) (hR : 0 < R)
  (x y : → ) ( : → ) (hm : Measurable ) (hm : Measurable )
  (hT : u Set.Icc (0 : ) b, | u - i range R, x i * u ^ i| K * u ^ R)
  (hT : u Set.Icc (0 : ) b, | u - i range R, y i * u ^ i| K * u ^ R)
  (hL : u Set.Icc (0 : ) b, | u - x 0| L * u) :
  K' : , N : , 1 N →
  |oneDScale b N h k
    - j range R, jetCoeff p k (chartJet x y R) j * N ^ (-(p + (j : ) / k))|
  K' * N ^ (-(p + (R : ) / k))
noncomputable def chartJetCoeff (x y : → ) (R j m : ) : :=
  range (j + 1), y * ((gPoly x R) ^ m).coeff (j - )
theorem jetCoeff_chartJet_eq ( p : ) (k : ) (h : 0 < ) (hp : 0 < p) (x y : → )
  (R j : ) :
  jetCoeff p k (chartJet x y R) j

```

= 1 / (k :) * m range R, ^ m / (m.factorial :) * chartJetCoeff x y R j m
 * weightedMass (x 0) (p + (j :) / k + m - 1)

2 Mathematics

`oneKernel_jet_expansion` (unit 92) turns a finite u -jet of an amplitude $F(u, s)$ with remainder $u^R H_R(s)$ into an expansion in $N^{-(p+j/k)}$, given moment integrability of $s^{p+j/k-1} e^{-\beta s^2} |F_j(s)|$ and of the remainder envelope. Unit 95 supplies, for $F = \eta e^{\beta s \xi}$, the jet $F_j = e^{\beta s x_0} P_j$ and $H_R(s) = C(1+s)^R e^{\beta s(x_0+L_1 b)}$. Two points remain and are settled here.

Moment integrability. The envelope $A(1+s)^R e^{\beta s a'}$ is expanded by the binomial theorem into $\sum_m A_m^{(R)} s^m e^{\beta s a'}$, and each term times $s^\gamma e^{-\beta s^2}$ is a shifted weighted kernel $s^{\gamma+m}$ `quadKernel`(β, a', s), integrable for $\gamma > -1$ by `logPowShift_weightedKernel_integrableOn`. The $(1+|\log s|)^j$ factor is carried along for later use with log-weighted terms although the present application has $j = 0$.

Coefficients. $P_j(s) = \sum_\ell y_\ell \sum_m \frac{(\beta s)^m}{m!} [u^{j-\ell}](g_R^m)$ is a polynomial in s , so $c_j = k^{-1} \int s^{p+j/k-1} e^{-\beta s^2} e^{\beta s x_0} P_j(s) ds$ integrates termwise (each term is a weighted kernel with exponent $p+j/k+m-1 > -1$) into the finite sum of $S_\lambda(x_0)$ values displayed in the abstract. In the paper's notation $S_\lambda(a) = \text{weightedMass } \beta a (\lambda - 1)$.

Domain of the envelope. The abstract transfer requires $H_R \geq 0$ on all of $\mathbb{R}$; the envelope is therefore defined with $(1+|s|)^R$ and reduced to $(1+s)^R$ on $s > 0$ where it is used.

3 Paper-facing meaning

This is the one-dimensional case of the all-order statement of `thm:TaylorTree` with an explicit remainder: the exponents $p+j/k$ are the poles of the tree, and the coefficients are finite combinations of $S_\lambda(x_0)$ with universal polynomial weights in β and the Taylor data (x_i, y_i) . The inverse- N series obtained by letting $R \rightarrow \infty$ is not claimed to converge; the theorem is a finite- R statement with an explicit $O(N^{-(p+R/k)})$ remainder, which is exactly the level of rigour of the paper's Lemma on the fluctuation expansion.

Follow-ups in the same line: (i) derive the Taylor-remainder hypotheses from C^R regularity via `taylor_mean_remainder_bound` (a C^∞ corollary with hypotheses only on ξ, η); (ii) the $d = 2$ block with noncritical face parameters (Astra #3 target 3); (iii) the weighted finite-part axis lemma (target 4).

4 Lean notes

- `rw [add_pow]` chose $(1+|\log s|)^j$ before $(s+1)^R$; pin the instance, `add_pow s 1 R`.
- `rw [← mul_assoc]` reassociated the wrong product in $s^\gamma \cdot (E \cdot (X \cdot \Sigma))$; pass the first factor, `← mul_assoc (Real.exp (- * s ^ 2))`.
- `positivity` cannot prove $0 \leq (1+s)^R$ for unconstrained real s ; define the envelope with $|s|$ when a global nonnegativity hypothesis is demanded.
- `Measurable.mul_const` takes the constant explicitly; `fun_prop` proves measurability of these exponential-polynomial expressions directly.
- `IntegrableOn.congr_fun` pointwise goals are beta-redexes: `beta_reduce` before `rw`.

- `MeasureTheory.integral_finset_sum` is deprecated for `integral_finsetSum`.